



SURAJ KIRAN AIRI

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EDUCATION

Chandigarh University

Bachelor of Engineering in Computer Science and Engineering, CGPA: 8.67

Mohali, India

Aug. 2022 – July 2026

EXPERIENCE

Freelance Gen AI

Mar. 2025 – Present

Self-employed (Remote)

- Built and deployed **3+ production GenAI applications** for client organizations, including RAG systems with document parsing, hybrid retrieval, reranking, and evaluation pipelines.
- Designed and integrated **agentic automation workflows** using LangGraph, LangChain, and LiteLLM, enabling tool execution, multi-step decision making, and workflow automation.

Flutter Developer Intern

Jul. 2023 – Sept. 2024

YoursThatSenior & SN Silos (2 paid internships)

Remote & Hybrid

TECHNICAL SKILLS

GenAI & Retrieval: RAG, Hybrid Search (Vector + BM25), Reranking, RRF, Metadata Filtering, RAGAS Evaluation

LLM Frameworks: LangChain, LangGraph, LiteLLM, MCP, LangSmith, Transformers

Backend: Python, FastAPI, Node.js, Express.js, REST APIs

Databases: PostgreSQL, MongoDB, FAISS, ChromaDB

Infrastructure & Tools: Docker, Linux, Git, Postman, Jupyter

PROJECTS

Zensho – AI Video Editing Pipeline | Python, FastAPI, LiteLLM, Deepgram, React, OpenTimelineIO, Electron Mar 2026

- Architected a **dual-environment system** with a Python/FastAPI native backend and React frontend, using IPC bridge for low-latency communication between inference layer and UI.
- Built an **end-to-end LLM inference pipeline** – Deepgram for sub-second audio transcription, LiteLLM for intelligent provider routing – to detect and slice silences, stutters, and bad takes with **word-level precision**.
- Designed **privacy-first BYOK architecture** (all media processed locally) with **OpenTimelineIO export** for direct handoff to DaVinci Resolve and Premiere Pro, keeping AI decisions fully auditable and user-correctable.

EduPath: Production RAG Chatbot | Python, FastAPI, LlamaCloud, Vector DB, BM25, RAGAS May 2026

- Built an **end-to-end RAG pipeline** for education-domain documents, using **LlamaCloud** parsing, structured **metadata extraction**, and source/date/category-based retrieval filtering.
- Implemented **hybrid retrieval combining dense vector search and BM25**, fused using **Reciprocal Rank Fusion (RRF)** and improved with cross-encoder **reranking** for higher retrieval relevance.
- Evaluated pipeline using **RAGAS**, achieving **context precision 0.87**, **context recall 0.89**, **faithfulness 0.93**, and **answer relevancy 0.90** on a domain-specific eval set.

MeroBot – Personal AI Assistant | Python, LiteLLM, LangGraph, AsyncIO, Telegram Bot API, SQLite Feb 2026

- Engineered a **production-grade agentic assistant** with iterative tool-calling loop (10 chained LLM calls), **structured outputs**, explicit fallback paths, sandboxed execution, and full conversation audit trail.
- Built **async message-bus agent orchestration** covering web search, code execution, file I/O, DB queries, and **sub-agent delegation** – designed for safe, extensible integration into domain-specific pipelines.
- Designed **pluggable LLM backend** (100+ providers via LiteLLM) with thread-safe tool registry, permissions model, and latency-optimized provider routing for reliable, provider-agnostic inference.

ACHIEVEMENTS

Hackathons: Winner of **Smart India Hackathon (SIH) 2024**, Software Edition.

Competitions: Ranked **443/3800+ (Top 12%)** in Kaggle ML Competition; Won Work Force 2023, IEEE CIS CUSB.

Open Source: Published **3 Python libraries on PyPI** – ai-tools-executor & visualkit.

Technical: Attended and taught **10+ ML sessions** at university; Multiple ML certifications.